What the pilot will test

A proposed three-month operating pilot in M4–M6, after the M3 launch gate. Definitions below operationalise the agreed implementation roadmap.[1]

PILOT AREA
1 township
Bounded area; site to be confirmed

PARTICIPATION BASIS
100 monthly active
Planning assumption; not a verified audience

REWARD VOLUME BASIS
200 / month
Expected physical water rewards; assumption

Agreed KPI	Target	What will be measured
Repeat participation	**≥30%** complete a second activity within seven days.[1]	Share of eligible first-activity accounts with a second distinct eligible activity within 168 hours; use a fully observed cohort. See page 2.
Reliable water rewards	**≥95%** successful redemption attempts; report cost per bottle dispensed.[1]	Confirmed physical dispensing with correct points accounting, divided by unique eligible redemption sessions. See page 3.
Useful planning evidence	Identify **three frequently used, low-shade path segments** for SDP review.[1]	Consented, aggregated route use paired with defensible shade evidence and explicit limitations. See page 4.

Volume assumptions and readiness

Budget basis: 100 active participants × four available rewards per month × 50% assumed uptake = 200 rewards/month; 600 over three full operating months. Four rewards and uptake are modelling assumptions, not measured demand or additional success KPIs.[2,3]

Launch gate — end of M3: compatible operator, confirmed water funding and successful end-to-end redemption tests.[1] Before resident launch, also verify hosted account persistence, handset recording and the reporting records described here.

Hackathon scope: this is a future pilot measurement plan. The current submission can demonstrate labelled prototype activity and simulated redemption; it does not require or claim completed resident research or pilot results.

Repeat participation: who counts?

Target: ≥30% complete a second activity within seven days.[1] Report the numerator and denominator alongside the percentage.

Seven-day repeat rate: within the cohort with complete 168-hour follow-up, divide accounts with a qualifying second activity by all accounts in that same cohort, then multiply by 100.

Definition	Proposed measurement rule
Eligible activity	An accepted, non-demo walking/running activity from an enrolled pilot account in the agreed area. Apply declared location-quality and plausibility checks; exclude test/admin records, duplicates, overlapping replays and rejected submissions. Checks do not prove authentic device coordinates.[2]
First and second activity	Use each account's first eligible pilot activity as its starting event. Count a second distinct, non-overlapping eligible activity completed after the first and no later than 168 hours after the first completion. Same-day activity can qualify under the agreed wording.
Complete observation window	At a report cut-off, include in the denominator only first activities that ended at least 168 hours earlier. Show newer accounts as awaiting follow-up, even when some already returned. Add the final cohort after a seven-day close-out window.
Monthly active participant	A distinct enrolled account with at least one eligible activity during the Malaysia calendar month. Planning, sign-in or a reward-only visit does not establish activity participation. Report accounts; unique people require separate enrolment controls.
Time and versions	Retain activity completion and server acceptance timestamps. Use elapsed hours for the seven-day metric and Malaysia time for calendar reporting. Keep activity/reward rule versions so changes remain visible.

Illustration only: 40 accounts have complete follow-up; 12 complete a qualifying second activity → 12/40 = 30%. Eight newer accounts remain outside this cohort until their full observation window ends.

Interpretation: establish early pilot baselines and compare weekly cohorts under the same definition. Record outages, organised events and rule changes. An observed return rate does not isolate the causal effect of water rewards.

Water reliability & operating cost

Target: ≥95% successful redemption attempts, with cost per bottle dispensed.[1] Live pilot results require physical operator confirmation.

Measure	Definition / reporting treatment
Eligible redemption session	One deliberate water-redemption request from an eligible account with sufficient points at an in-scope selected machine. Give the logical session a stable ID from the request stage; repeated clicks, scans or network retries remain that session.
Success rate	Unique sessions with confirmed physical dispensing and exactly one correct points deduction ÷ all eligible sessions initiated in the reporting period × 100. A voucher alone or a simulated success is not physical fulfilment.
Failures and uncertainty	Include legitimate eligible requests that fail during voucher creation, communication, stock fulfilment or accounting. Unresolved sessions remain in the denominator without counting as successes at the cut-off. Separately log invalid, unauthorised and duplicate requests.
Reconciliation	Match session, voucher, machine, operator confirmation and points ledger. If a dispensing outcome is unknown, reconcile before refund or another dispense. Preserve the report cut-off and record later resolutions as traceable adjustments.
Direct cost per bottle	Actual bottled-water fulfilment charges for the period ÷ confirmed bottles dispensed. Identify which operator fees are included; report separate refill units if that water type is used.
Programme cost per bottle	Actual recurring programme costs for the period ÷ confirmed bottles dispensed. Show setup expenditure separately. Use A4 as the planning baseline and explain actual cost/volume differences.[3]

Illustration only: 100 eligible sessions; 95 successful, three failed and two unresolved → 95/100 = 95%. Disclose all statuses. If there are no eligible sessions or confirmed bottles, the associated rate or unit cost is **not assessable**, never an assumed zero or 100%.

Operational companion measures: show unresolved count/age, stock outages, points discrepancies, unique redeemers and redemption volume against the funded budget. Agree response timeouts and responsibility for exceptions with the operator before launch.

Planning evidence: use, shade & consent

Target: three frequently used, low-shade path segments for SDP review.[1] This is a candidate-review output, not proof of a completed infrastructure improvement.

Step	Proposed method and evidence requirement
1. Use consented activity	Include accepted non-demo activities with permission for planning insights. Map valid traversals to stable path-segment IDs; do not bridge location gaps as observed walking. Record matching uncertainty and coverage.
2. Aggregate and rank use	Rank eligible segments by distinct contributing accounts; retain traversal counts as context. Propose a minimum of five distinct consenting accounts per displayed segment over the pilot, subject to SDP review. Suppress smaller groups and precise home/start locations; this threshold alone does not guarantee anonymity.
3. Assess shade evidence	Agree shade criteria and departure-time bands before scoring. Use suitable attributed public height, canopy or covered-way inputs, recording source date, method and limitations. An assumed-height building scenario does not establish actual shade coverage.[4]
4. Prepare three candidates	For each: segment location, aggregated use, relevant time band, shade evidence/coverage, uncertainty and a question for SDP assessment. Record SDP's review response; recommendations remain subject to site assessment.

Current data gap: the bounded OSM snapshot contains no building-height tags or tree-canopy features.[4] If defensible shade inputs remain unavailable, report frequently used segments as **candidates with unverified shade**. Mark the low-shade KPI not assessable; do not count assumed scenarios as a verified result.

Minimum proposed reporting records

Activity: pseudonymous account, activity ID, source, completion time, accepted distance and validation/rule version. **Redemption:** session/voucher/machine IDs, timestamps, outcome, ledger reference and cost evidence. **Planning:** consent record, segment aggregates and data/method provenance. Add operator outcomes and reporting events where the prototype does not yet capture them.[2]

Restrict raw movement access; agree retention, deletion and withdrawal handling before launch. Publish aggregate reports without voucher credentials, identifiable trails or small-group detail.

Review cadence & expansion decisions

Use the agreed roadmap for timing.[1] The decision rules below are proposed review guidance; targets are team assumptions, not industry benchmarks.

When	Proposed owner	Review / output
Before M4 launch	Pilot lead + technical team + operator + SDP	Confirm launch gate, funding, definitions, event logging, consent controls, route area and operator exception handling. Lock the measurement version.
Weekly in M4–M6	Pilot lead + technical team + operator	Review eligible activity cohorts, eligible sessions, failures/unknown outcomes, wallet discrepancies, stock, spend and data completeness. Preserve original counts.
Monthly in M4–M6	Project team + SDP	Report monthly active accounts, fully observed repeat cohorts, redemption success, unit costs and aggregate planning candidates. Establish baselines and explain incidents or rule changes.
M6 close-out + 7 days	Project team + SDP + operator	Complete follow-up for final first-activity accounts; report the same cut-off for redemption outcomes and unresolved cases. Decide improvements for M7–M9.
M10–M12 review	SDP + delivery partners	Assess expansion only with evidence, operator readiness and affordable water funding. Each added location needs its own data and machine readiness review.

Decision flags

PROCEED TO EXPANSION REVIEW: targets supported by usable evidence, reliable accounting, confirmed delivery capacity and funding. **REFINE / EXTEND:** missed targets, insufficient sample/coverage or missing shade evidence; resolve the cause and repeat measurement. **PAUSE AFFECTED FLOW:** unresolved duplicate spending, privacy incident, unsafe operation or unfunded water obligations.

Basis and reference register

1. Agreed Core 8 — Implementation Roadmap.
StrideEco proposal baseline: schedule, 100 monthly active participants, 200 redemptions/month, three KPIs and M3 launch gate.

2. StrideEco prototype records.
lib/pilot.ts; docs/product-brief.md; docs/account-loading.md. Current prototype rules and data concepts; physical dispensing and resident-pilot outcomes remain unestablished.

3. StrideEco Cost Breakdown — A4.
Current document version 1, read on 23 October 2026; planning model dated 22 October. Volume and cost estimates remain assumptions and allowances.

4. Public-data audit.
docs/data-sources.md; public/data/map-provenance.json. Bounded Elmina OSM snapshot retrieved 22 October 2026; missing height/canopy inputs and scenario limits.